

Ice Cream Parlour Management System (ICPMS)

Database Architecture & UI/UX Design System

Version: 1.0

Database:

- MongoDB Atlas
- Mongoose ODM

UI Framework:

- React
 - Tailwind CSS
-

PART 1: DATABASE ARCHITECTURE

1. Database Design Principles

The database should:

- Support business growth
 - Support analytics
 - Avoid data duplication
 - Maintain relationships
 - Provide fast queries
 - Support future multi-branch expansion
-

2. Database Collections

System Collections:

users

products

categories

inventory

customers

orders

orderItems

invoices

employees

offers

notifications

attendance

salesReports

settings

3. Users Collection

Purpose:

Authentication & Authorization

Fields:

_id

name

email

password

role

phone

profileImage

status

lastLogin

createdAt

updatedAt

Role Values:

admin

manager

cashier

inventory_staff

Indexes:

email (unique)

phone

4. Categories Collection

Purpose:

Product Categorization

Fields:

_id

name

description

status

createdAt

updatedAt

Examples:

Cone

Cup

Shake

Sundae

Family Pack

Beverage

5. Products Collection

Purpose:

Store Ice Cream Products

Fields:

_id

name

description

categoryId

price

costPrice

taxPercentage

stock

imageUrl

cloudinaryPublicId

status

createdBy

createdAt

updatedAt

Indexes:

name

categoryId

status

6. Inventory Collection

Purpose:

Raw Material Tracking

Fields:

_id

itemName

quantity

unit

minimumQuantity

supplierName

purchasePrice

lastUpdated

status

createdAt

updatedAt

Examples:

Milk

Chocolate Syrup

Cone

Cup

Dry Fruits

Chocolate Chips

7. Customers Collection

Purpose:

Customer Records

Fields:

_id

name

mobile

email

address

birthday

loyaltyPoints

totalOrders

totalSpent

createdAt

updatedAt

Indexes:

mobile

email

name

8. Employees Collection

Purpose:

Employee Management

Fields:

_id

employeeCode

name

email

phone

role

salary

address

joiningDate

status

createdAt

updatedAt

Indexes:

employeeCode

email

phone

9. Orders Collection

Purpose:

Store Sales Orders

Fields:

_id

orderNumber

customerId

cashierId

items

subtotal

discountAmount

taxAmount

totalAmount

paymentMethod

paymentStatus

orderStatus

invoiceId

createdAt

updatedAt

Payment Methods:

Cash

UPI

Card

Order Status:

Pending

Preparing

Ready

Completed

Cancelled

Indexes:

orderNumber

customerId

createdAt

10. Order Items Structure

Embedded Inside Orders

Fields:

productId

productName

quantity

unitPrice

totalPrice

11. Invoices Collection

Purpose:

Invoice Records

Fields:

_id

invoiceNumber

orderId

customerId

totalAmount

taxAmount

discountAmount

pdfUrl

generatedAt

Indexes:

invoiceNumber

orderId

12. Offers Collection

Purpose:

Discount Management

Fields:

_id

offerName

description

discountType

discountValue

applicableProducts

startDate

endDate

status

createdAt

Discount Types:

Percentage

Flat

BOGO

Combo

13. Notifications Collection

Purpose:

System Alerts

Fields:

_id

title

message

type

isRead

userId

createdAt

Types:

Low Stock

Order

Invoice

Offer

System

14. Attendance Collection

Purpose:

Employee Attendance

Fields:

_id

employeeId

date

checkIn

checkOut

workingHours

status

Status:

Present

Absent

Half Day

Leave

15. Settings Collection

Purpose:

Store Configuration

Fields:

_id

storeName

gstNumber

address

phone

email

invoicePrefix

taxPercentage

updatedAt

16. Collection Relationships

Users | — Products — Orders — Inventory — Reports

Customers | — Orders

Orders | — Order Items — Invoice — Payments

Products | — Categories

17. Indexing Strategy

Indexes Required:

users.email

customers.mobile

products.name

orders.orderNumber

invoices.invoiceNumber

employees.employeeCode

notifications.userId

Purpose:

Fast Search

Fast Reports

Fast Dashboard Loading

18. Analytics Aggregations

Dashboard Metrics:

Today's Sales

Monthly Sales

Today's Orders

Revenue

Top Products

Top Customers

Low Stock Items

Monthly Growth

Required MongoDB Aggregations:

\$group

\$lookup

\$match

\$project

\$sort

\$sum

\$count

PART 2: UI/UX DESIGN SYSTEM

19. Design Philosophy

Style:

Modern SaaS Dashboard

Keywords:

Professional

Clean

Minimal

Business Focused

Fast

Responsive

Similar Inspiration:

POS Systems

Restaurant ERP

Retail Management Software

20. Color Palette

Primary

2563EB

Blue

Secondary

1E293B

Dark Slate

Success

16A34A

Green

Warning

F59E0B

Orange

Danger

DC2626

Red

Background

F8FAFC

Card Background

FFFFFF

Border

E2E8F0

Text Primary

0F172A

Text Secondary

64748B

21. Typography

Font Family:

Inter

Headings:

Bold

600–700

Body:

400–500

Hierarchy:

H1

32px

H2

24px

H3

20px

Body

16px

Caption

14px

22. Layout Structure

Navbar

↓

Sidebar

↓

Content Area

↓

Footer

23. Dashboard Layout

Top Section:

6 KPI Cards

Today's Revenue

Today's Orders

Customers

Products

Inventory

Low Stock

Middle Section:

Sales Chart

Revenue Chart

Bottom Section:

Recent Orders

Low Stock Table

Top Products

24. Product Module UI

Listing Page:

Search Bar

Filters

Product Table

Pagination

Actions

Add Product Button

Product Card View

Product Table View

25. POS Billing Screen

Layout:

Left:

Product Search

Category Filter

Product Grid

Right:

Cart

Summary

Payment Section

Invoice Preview

Optimized For:

Touch Screen Usage

Fast Billing

Minimal Clicks

26. Customer Module UI

Customer Table

Search

Profile Drawer

Purchase History

Loyalty Section

Analytics Cards

27. Reports UI

Date Filters

Charts

Tables

Export Buttons

PDF Download

Excel Download

Print Option

28. Responsive Design Rules

Mobile First

Breakpoints:

640px

768px

1024px

1280px

1536px

Requirements:

Sidebar Collapse

Responsive Tables

Mobile Cards

Adaptive Charts

Touch Friendly Inputs

29. Reusable Components

Button

Input

Textarea

Select

Checkbox

Radio

Table

Modal

Drawer

Tooltip

Badge

Card

Chart

Avatar

Loader

Toast

Pagination

File Upload

Confirmation Dialog

30. UX Requirements

Fast Navigation

Clear Feedback

Loading States

Skeleton Loaders

Success Messages

Error Messages

Confirmation Modals

Form Validation

Accessibility Support

Keyboard Navigation

31. Cloudinary Image UX

Image Preview

Drag & Drop Upload

Image Replace

Image Remove

Upload Progress Indicator

Error Handling

32. Production Ready Standards

Consistent Spacing

Consistent Colors

Consistent Typography

Reusable Components

Accessibility Compliance

Responsive Design

Optimized Performance

Clean User Experience

Final Deliverables

Database:

- ✓ MongoDB Architecture
- ✓ Collection Relationships
- ✓ Index Strategy
- ✓ Analytics Aggregations
- ✓ Future Scalability

UI/UX:

- ✓ Design System
- ✓ Dashboard Layout
- ✓ POS Layout
- ✓ Mobile Responsive Rules

✓ Component Library

✓ Professional SaaS Experience

End of Document